Uncertainty-arbitrated action selection for emotion-driven artificial creatures

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Abstract

In the dl2l artificial-life simulator, adding episodic memory to action selection made creatures worse: they ate up to 3.6× less and died 27% sooner than creatures without memory. The cause is architectural. Memory, the operant-conditioning table and, more recently, a learned world model are chained in a fixed order, so whichever filter runs first decides, regardless of how much it knows. We propose replacing that chain with uncertainty-based arbitration, a principle from computational neuroscience (Daw, Niv & Dayan 2005). Each controller reports, for every candidate action, an estimated value and how unsure it is of that estimate. Cheap controllers are consulted first, and a more expensive one is consulted only when the cheaper ones cannot tell the best action from the runner-up and the creature's arousal leaves room to deliberate. All controllers value outcomes in one unit, the reduction of the creature's drive deficits, and episodes are delimited by attentional focus on one drive rather than by a game's end. Exploration is directed by uncertainty through Thompson sampling. In a pre-registered simulation on 200 abstract environments, the arbiter regulated drives better than episodic memory alone and the planner alone, and as well as the best single controller, without being told which one that is. In a food–bed corridor, the episode marker eliminated dithering between drives. We describe how to integrate the mechanism into dl2l, its risks with the JEPA world model, and how to validate it against animal behaviour rather than survival alone.

1 Introduction

dl2l is a distributed artificial-life simulator in which creatures with drives (hunger, sleep), affects (pain, tedium), neuromodulators and hormones forage in a shared world. It descends from two earlier architectures: Campos et al. (2015), who modelled a minimal emotion-driven nervous system, and Mapa (2009), who added an experiential memory of emotionally valued episodes. Both reported large survival gains from memory: 6.7× in Campos et al. and about 2.5× in Mapa.

Reproducing those gains in dl2l failed. A behaviour-parity campaign (issue #84) found the opposite effect: creatures with memory ate 3.6–65× less and died 27% sooner (issue #88). Diagnosis showed that the memory filter ranked actions by how often they had been taken rather than by how well they had worked. Fixing that (PR #89) and letting memory choose the object while the operant table chooses the action (PR #92) raised feeding in every comparison, but the survival effect remained small (1.11× against Campos's 6.7×). Two problems remain:

1. **Arbitration is positional.** Filters run in a fixed order and the first one that leaves a single candidate decides. Memory acts on one or two engrams early in life and cannot be overruled later. The JEPA world model, added as a further filter, only runs when the earlier ones fail to decide.

2. **Exploration happens only by accident.** A creature explores when memory has nothing to say. Once memory holds anything, it exploits, even when what it holds is little evidence.

This document proposes a replacement for the action-selection chain. Its idea is simple: each controller should influence behaviour in proportion to how reliable it is in the current situation, and expensive deliberation should be used only when it is likely to change the choice and the creature can afford it. Concretely, we contribute:

- a common value for all controllers, the reduction of the creature's drive deficits, so their estimates can be compared;
- an episode marker for a world without a "game over": attentional focus on one drive, with hysteresis;
- an engram key and similarity metric built by hand from percepts and internal state, so episodic memory needs no trained encoder;
- a value and an uncertainty estimate for each controller (operant table, episodic memory, world model), and an arbitration rule that combines them under a budget set by behavioural efficiency;
- a simulation study in two abstract environments, including a pre-registered test on 200 environments;
- an integration plan for dl2l (Appendix A), its risks (Appendix B), a validation plan based on behavioural signatures known from animal studies, and a correspondence between the mechanism without planning and the active inference framework (Appendix C).

Structure of the document. Section 2 describes action selection in dl2l and in the Campos and Mapa architectures. Section 3 reviews the neuroscience of multiple controllers and episodic control in reinforcement learning, including our reproduction of two episodic-control algorithms. Section 4 presents the proposed mechanism and Section 5 compares it with Mapa and Campos. Sections 6 and 7 report the simulation study. Section 8 gives the validation plan and Section 9 concludes. Appendix A describes the integration into the dl2l code, Appendix B the risks, and Appendix C the correspondence between the mechanism and the active inference framework.

2 Action selection in dl2l and its predecessors

2.1 Campos et al. (2015)

Campos et al. model a minimal nervous system as concurrent, asynchronous components (perception, partial and full appraisal, homeostatic regulation, valuation, action selection). Each drive has a deprivation level; the drive with the highest deprivation is the one the creature attends to. Three ideas from this model matter here.

Behavioural efficiency. Following Hebb, efficiency E depends on arousal A (the deprivation of the attended drive) through an inverted U:

$$E(A) = -aA^2 + bA + c, \qquad a = 40/49,\ b = 40/7,\ c = 0$$

With these values E rises from 0 at A = 0 to its maximum of 10 at A = 3.5 and falls back to 0 at A = 7, the lethal level. In Campos et al. efficiency only sets the speed of movement. We will use it to set how much deliberation the creature can afford (Section 4.5).

Action selection as a chain of filters. Candidate actions are the affordances of the current situation (approach, eat, avoid, rest, wander), intersected with the action tendency of the attended

drive (hunger favours approach and eat). If more than one remains, only the nearest instance of each kind is kept. If still more than one remains, long-term memory chooses among them with probability proportional to remembered value; if memory holds nothing about the object, the choice is random.

Memory as accumulated outcome. Each fruit type has a weight in long-term memory, updated after eating as $W_i = W_{i-1} + Q_i$, where Q is the fruit's nutritional value. Actions with negative remembered value are never chosen again. In their experiments memory was not used for roughly the first 150 interactions, after which it replaced random choice.

2.2 Mapa (2009)

Mapa adds an experiential memory organised in episodes. The unit of experience is the Micro Emotional Experience (MEE): a stimulus, the action taken and the resulting emotional valence. When the creature is dysregulated it may need several actions to recover; the ordered sequence of MEEs that restores homeostasis is a Complete Experience with Qualia (CEQ), the unit of long-term memory. Each CEQ has an emotional intensity (IMR) that grows when the memory is evoked and works again and decays when it does not. How much a CEQ consolidates depends on arousal through an inverted U, and extreme arousal creates flashbulb memories of maximum intensity that do not fade.

In Mapa's action selection, full appraisal first selects the stimulus with the highest expectancy, that is, the highest expected emotional value according to the evoked memories, and operant conditioning then selects the action towards it. Mapa reports average survival rising from 1,850 to 4,700 cycles with memory.

2.3 The current dl2l code

dl2l keeps the filter chain of Campos et al., adds Mapa's division of labour, and adds a learned world model. On every cognitive cycle `FullAppraisal` runs, in this order: the innate action-tendency filter, `TargetDistanceFilter` (nearest instance), `MemoryFilter`, `ActionProbabilityFilter` (the operant table), `WorldModelFilter` (JEPA) and `RandomFilter`. The chain stops at the first filter that leaves one candidate.

Component	What it does today	Limitation
`MemoryFilter`	Scores each visible object type by the mean of (−emotion change × eligibility) over recent consummatory engrams, then samples an object in proportion to the score, with a novelty prior modulated by tonic dopamine	Not conditioned on the current situation or internal state; no notion of how much evidence it has
Operant table	Action probabilities per object type, nudged up or down after each outcome; tonic dopamine flattens them (softmax temperature)	Keyed by object type only; no estimate of its own reliability
`WorldModelFilter`	Scores each candidate by the JEPA's predicted aversive cost (pain + fear), one step ahead, only when arousal ≥ 4.5, and disables itself when latent prediction error is high	Blind to hunger and sleep outcomes; reached only when earlier filters fail to decide
`Engram`	Stores the action, the perception, the dominant emotion at decision time, a scalar emotion change and an eligibility trace	No internal-state vector and no per-drive outcome

The expectancy of Mapa exists in dl2l in two places: as the object score of `MemoryFilter`, and as the expectancy predictor of the neuromodulation loop (issue #57), which turns expectation errors into dopamine. Neither is conditioned on the creature's internal state, which is what Section 4 changes.

3 Multiple controllers and episodic control

3.1 Three controllers in the brain

Animals choose actions with more than one system. A habitual system caches the value of actions in situations and responds fast but adapts slowly; it is associated with the dorsolateral striatum. A goal-directed system simulates consequences with a model of the world; it adapts quickly but is slow and effortful; it is associated with prefrontal cortex. Daw, Niv & Dayan (2005) proposed that the two compete according to the uncertainty of their predictions, so that each controls behaviour when it is likely to be the more accurate. Lee, Shimojo & O'Doherty (2014) found reliability signals for both systems in lateral prefrontal cortex. Keramati, Dezfouli & Piray (2011) added the time cost of deliberation, and Pezzulo, Rigoli & Chersi (2013) proposed that simulation is used only when its value of information exceeds its cost.

Lengyel & Dayan (2007), *Hippocampal contributions to control: the third way*, added a third controller: episodic memory, which repeats what worked on a past occasion without averaging. Their analysis shows that episodic control beats a model-based controller with computational noise when experience is scarce, and that the model-based controller overtakes it later. We replicated their tree-structured environment and controllers. The replication reproduced the main result: episodic control wins first, and the crossover comes later as the environment grows more complex (after iteration 2, 4 and 6 for branching factors 2, 3 and 4). That replication is the basis of the simulation study in Section 6.

3.2 Episodic control in reinforcement learning

Two algorithms brought episodic control into deep reinforcement learning. Model-Free Episodic Control (MFEC; Blundell et al. 2016) keeps, for each action, a table from state embeddings to the best return ever obtained after taking that action. Unseen states are valued by averaging their k nearest neighbours. The embedding is fixed: a random projection or a pretrained auto-encoder. Neural Episodic Control (NEC; Pritzel et al. 2017) replaces the fixed embedding with one learned end to end through a differentiable nearest-neighbour lookup, and reports better data efficiency than DQN early in training.

3.3 Our reproduction of MFEC and NEC (snake_rl)

Why. Our earlier review of the dl2l memory problem recommended an episodic controller in the style of MFEC or NEC. Before building one into dl2l, where every experiment takes hours on a cluster, we wanted to know which of the two holds up and under what conditions. NEC's appeal was that it learns its own representation, which would pair naturally with a JEPA encoder.

What. We wrote pure-NumPy implementations of DQN, N-step DQN, MFEC, episodic control on a frozen random embedding, and NEC, with hand-derived gradients checked against finite differences. We tested them on our own Snake game (boards from 7×7 to 25×25, with obstacles, bonus food, distractor noise channels, and Atari-style 84×84 pixels with DQN's network) and on a Ms. Pac-Man clone. The study ran 12 experiments, 4 of them pre-registered, and closed on 12 October 2026 (summary). A final run of 1M steps is still in progress on the cluster.

What we found.

Finding	Evidence	Consequence for dl2l
MFEC works where states repeat	Best and fastest agent on clean 7×7 Snake (1.58 fruit per episode vs DQN 1.04, NEC 1.01); a quarter of lookups hit an exactly stored state	dl2l's abstract percepts repeat often, which suits MFEC
MFEC fails when states stop repeating	With 4 noise channels: 0% exact hits, 0.23 vs random 0.21	Dimensions of the memory key that do not predict value act like noise channels
NEC's learned embedding rarely helped	At 250k steps every NEC arm was 0.22–0.31 below a frozen random embedding	Do not train an encoder through the memory lookup
NEC is sensitive to input magnitude	A constant input at 1.0 made the embedding diverge in 3 of 3 seeds; at 0.235, in 0 of 3	Unscaled features such as raw distance must be normalised
NEC is sensitive to reward magnitude	Ms. Pac-Man NEC 357 in small reward units vs 519 in Atari units; MFEC 390 vs 385	Prefer estimators whose behaviour does not depend on the scale of the outcome
The representation decides, not the memory	Raw pixels were a better neighbour space than any trained encoder (position purity 0.78 vs 0.43 for NEC)	A hand-designed key can work; a learned latent is an upgrade, not a prerequisite
NEC's published advantage did not survive equal tuning	NEC led a paper-style DQN on Ms. Pac-Man (519 vs 478) until one DQN setting was tuned (668)	No part of our design rests on NEC's data-efficiency claim

The conclusion is to use MFEC-style episodic memory, on a key designed for dl2l, and not NEC. The limits are that both games are our own, budgets were small (at most 1M steps) and several sweeps used 2–5 seeds.

4 The proposed mechanism

The mechanism replaces the memory, operant and world-model filters of the current chain with one arbiter. The innate filters that come before them (action tendency, nearest target) are kept. On every cognitive cycle:

1. the candidates are the (action, target) pairs left by the innate filters;
2. each controller that the creature can afford at its current arousal estimates, for every candidate, its value and the uncertainty of that value;
3. controllers are consulted from cheapest to most expensive, and the consultation stops as soon as one is decisive, that is, confident that its best candidate beats the runner-up;
4. one candidate is chosen by sampling from the estimates of the most decisive controller, so that uncertain options still get tried.

Figure 1 shows the flow. The subsections define each part: the value all controllers estimate (4.1), how experience is cut into episodes (4.2), how episodic memory recognises similar situations (4.3), how each controller estimates value and uncertainty (4.4), and how the arbiter decides and explores (4.5).

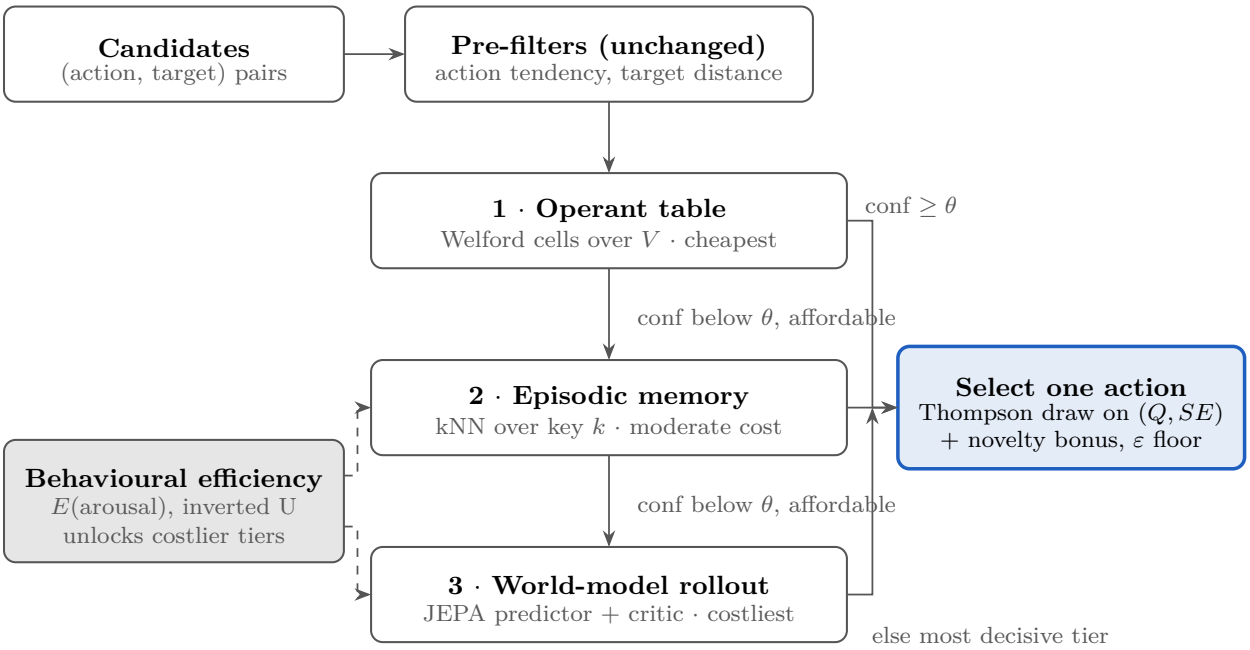


Figure 1. arbitration between three controllers under an efficiency budget.

4.1 A common value: drive reduction

To compare controllers, they must estimate the same quantity. We use the reduction in the creature's homeostatic deficit, as in homeostatic reinforcement learning (Keramati & Gutkin 2014).

$$d(H) = \left\| w \odot \max(0,\, H - H^*) \right\|_2 \qquad V(K \mid H) = d(H) - d(H + K)$$

The symbols are:

- **H**, the internal state: the vector of current drive and affect levels (hunger, sleep, pain, tedium), each between 0 and the lethal level 7.
- **H***, the set point: the level each drive is regulated towards. We propose the lower edge of the homeostatic equilibrium band of issue #59, 0.18.
- **max(0, H − H*)**, the deficit: how far each drive is above its set point. Drives below their set point count as zero.
- **w**, precision weights, one per drive, 1 by default; ⊙ multiplies element by element. They state how much each deficit matters (see below).
- **d(H)**, the overall deficit: the length of the weighted deficit vector. It is 0 when every drive is at or below its set point and grows as any drive rises.
- **K**, the outcome: the change in drive levels that an action caused. Eating a green apple might give K = (−2.4, 0, 0, 0): hunger falls by 2.4, the rest is unchanged. Engrams store K, not a scalar valence.
- **V(K | H)**, the value of outcome K for a creature in state H: how much the outcome reduces the overall deficit. Positive values are good.

The same outcome is worth different amounts in different states, which is what a hungry or sleepy

animal shows. With two drives, an apple with K = (−2, 0) is worth V = 1.95 to a creature at H = (5, 1), which is hungry, and only V = 0.07 to one at H = (1, 5), which needs sleep. An outcome that relieves an irrelevant drive is worth almost nothing, and relief is worth more the larger the deficit. This also removes the failure of issue #91, where engrams were tagged with whatever drive dominated at decision time: K is recorded independently of need, and the need is applied when the memory is used.

The weights w are where emotion enters. The central hypothesis of the dl2l thesis is that composite emotions act as precision: they change how much each drive matters rather than what the creature knows (after Pezzulo, Rigoli & Friston 2018 and Hesp et al. 2021). Here that hypothesis has a concrete place. Stress could raise the weight of pain; apathy could flatten all weights. Memories need not be relearned, because only V changes, not K. In this proposal w = 1, and emotion-modulated w is the experiment that follows.

4.2 Episodes: attentional focus on one drive

Episodic memory needs episodes: a value is attached to a step by looking at what followed it until the episode ended. In Snake or Atari an episode ends when the game is won or lost. A creature's life has no such marker, so we define one: **an episode is one bout of attentional focus on a single drive.** This is the operational form of Mapa's CEQ, an ordered sequence of steps that ends when the attended emotion is regulated.

Urgency. Drives live on different scales, so focus compares a normalised urgency rather than raw levels:

$$u_i = w_i \cdot \frac{\max(0,\, H_i - H_i^*)}{L_i - H_i^*}$$

Here H_i is the level of drive i, H_i^* its set point, L_i its lethal level (7 for hunger and sleep; for affects that cannot kill, their maximum) and w_i its precision weight. Urgency is 0 at the set point and 1 at the lethal level.

The focus rule uses hysteresis, so the creature commits to a drive instead of switching every cycle:

Threshold	Meaning	Value
θ_{on}	A drive this high needs attention and may open an episode	level 2.0, the upper edge of the #59 band
θ_{off}	The focal drive is regulated; the episode closes as a success	level 0.5
Δ	Another drive takes the focus only if its urgency exceeds the focal drive's by Δ	0.3 in urgency units
θ_{crit}	Emergency: another drive this high takes the focus at once, but only while the focal drive is below θ_{crit}	level 4.0, and any pain that triggers escape
T_{max}	The focal drive has not fallen for this many cycles: the episode is abandoned	3 × the median steps-to-outcome for that drive

θ_{off}, Δ and θ_{crit} were set from the corridor simulations in Section 7.3: a small Δ made creatures dither to death, and an emergency threshold at 5.5 fired too late to reach the resource.

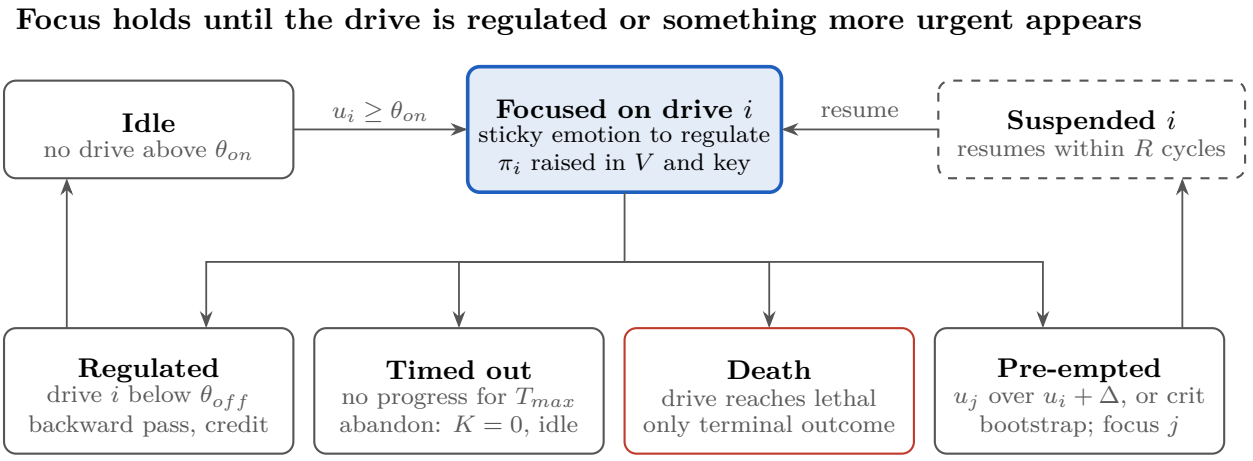


Figure 2. the attentional focus that delimits an episode.

How an episode closes, and what each step learns. The reward of step t is the drive reduction of whatever happened at that step, and each engram stores the discounted sum of the outcome vectors that followed it until the episode closed:

$$G_t = \sum_{j=0}^{T_e - t} \gamma^j K_{t+j}, \qquad \text{value at retrieval: } V(G_t \mid H_{\text{now}})$$

T_e is the step at which the episode closed and $\gamma = 2\hat{}(-1/5) \approx 0.87$, the decay already used by dl2l's eligibility traces (half-life of 5 cycles). Storing the vector G rather than a scalar return lets the same engram be valued under any future need.

Two stores. The episode defines a short-term and a long-term store. While an episode is open, its engrams (key, action, time) wait in an episode buffer and carry no value yet. When the episode closes, each engram receives its return G and is written to the per-action store M_a, the long-term memory that retrieval searches. Closing an episode is therefore the moment of consolidation, as in Mapa's model, where the micro emotional experiences (MEE) of an episode are grouped into a complete experience (CEQ) once homeostasis is restored. Her flashbulb memories correspond to a write that does not wait for the episode to close, which we leave for later. The buffer is not queried when choosing an action, so we call it an episode buffer rather than working memory. Letting decisions read the open episode, for instance to avoid repeating an action that has already failed in it, would make it a working memory; we have not tested this.

- **Regulated:** G is computed back from the end of the episode. This is MFEC's backward pass: a Monte Carlo return to the end of the episode. Unlike MFEC, which keeps the best return ever seen, we average, because dl2l's outcomes are noisy and a maximum locks onto lucky outcomes.
- **Pre-empted:** the sum is cut where focus moved and completed with the controllers' current estimate for that state. This is the bootstrapped N-step return of NEC. Pre-emption is not a failure, and the episode is suspended: if focus returns to the same drive within R cycles, it resumes.
- **Abandoned (timeout):** approach steps receive G = 0, so actions that led nowhere stop gaining credit.

- **Death:** the only terminal state; the focal episode closes with the lethal deviation as its outcome.

During an episode the focal drive replaces the per-cycle most aroused emotion as the emotion to regulate, generalising the anti-flicker gate that dl2l already applies to sleep (`MIN_SLEEP_TICKS`). Its precision weight is raised, so memories formed under the same focus are retrieved more readily.

4.3 Engram key, encoder and similarity

Episodic memory values a new situation by looking up the engrams of similar past situations. "Similar" needs a representation of the situation and a distance on it. MFEC obtains its representation from a fixed random projection or a pretrained auto-encoder; NEC learns one. The classic version of dl2l has neither: only the JEPA version has an encoder. We therefore propose a **hand-built, untrained encoder**: a fixed function φ that turns the creature's percept and internal state into a vector. The snake_rl study supports this choice: a learned embedding was rarely better than a fixed one, and training an encoder through the memory lookup was the fragile part of NEC. A fixed encoder also cannot drift, so stored keys remain valid for the creature's whole life.

The key of an engram has two blocks:

$$k = \varphi(p, h) = \left[\ \tfrac{1}{\sqrt{n_p}}\, z(p) \ \ ; \ \ \tfrac{\lambda}{\sqrt{n_h}}\, z(h) \ \right]$$

- **p, the percept**, in the creature's own frame: one-hot object type, distance divided by the vision radius, sine and cosine of the bearing, and a contact flag.
- **h, the internal state**: drive levels (`ht_*`), tonic dopamine, serotonin and orexin (`nm_*`) and cortisol (`end_*`), the same features `FullAppraisal` already builds for the JEPA.
- $\mathbf{z(\cdot)}$ standardises each feature (subtract its population mean, divide by its standard deviation), using the statistics dl2l already ships in `model_contract.json`. Without it, a large-valued feature such as raw distance would dominate every distance, which is the magnitude problem snake_rl exposed.
- $\mathbf{n_p}$ **and** $\mathbf{n_h}$ are the number of features in each block. Dividing by their square roots makes both blocks contribute equally to a distance regardless of how many features they hold.
- λ sets how much the internal state counts relative to the percept when deciding which memories are similar. At $\lambda = 0$, retrieval ignores how the creature feels and matches only what it sees; at $\lambda = 1$, both count equally. λ changes which memories are retrieved, never how they are valued: a retrieved outcome is always valued against the current state, $\mathrm{V(G \mid H_{now})}$. We use $\lambda = 0.5$, the best value in the simulations (Section 7.2).

Distance and neighbour weights. For a query key k we find, separately for each candidate action, the m nearest stored keys (m = 10) under the weighted Euclidean distance

$$D(k, k_j) = \sqrt{\textstyle\sum_i \pi_i \,(k_i - k_{j,i})^2}, \qquad \omega_j = \exp(-D(k, k_j)^2 / 2b^2)$$

π_i are per-feature precision weights (1 by default, the place where emotion could modulate retrieval). Each neighbour j receives a weight ω_j that falls with distance; the bandwidth b is the median distance of the m neighbours, so the weighting adapts to how densely memory covers the query. MFEC itself averages its m neighbours with equal weights. We weight them because the weights tell us how many neighbours are really close (Section 4.4); MFEC's equal weighting remains an ablation.

4.4 What each controller reports

For every candidate action a, each controller reports three numbers, all in units of drive reduction:

- $\mathbf{Q_a}$, the estimated value: the expected V of taking a now.
- $\mathbf{SE_a}$, the standard error: how unsure the controller is about Q_a itself. It shrinks as evidence accumulates.
- $\sigma_\mathbf{a}$, the outcome spread: how much single outcomes vary around Q_a, even when Q_a is known exactly. It does not shrink with evidence: a fruit tree may sometimes be empty.

The distinction matters. SE decides whether a controller knows enough to choose. σ is needed to check, after the fact, whether a controller's SE was honest (calibration, below).

Operant table (habit). Cells are keyed by (focal drive, object type, distance bin). Each cell keeps, per action, a count n, a running mean and a running variance of the values V experienced, updated in constant time with Welford's algorithm. Older evidence fades: on each update the count is multiplied by 0.995. Then

$$Q_a = \frac{n\,\bar{v} + \kappa\,Q_0}{n + \kappa}, \qquad \sigma_a = s, \qquad SE_a = \frac{\sigma_a}{\sqrt{n + \kappa}}$$

$\bar{v}$ is the running mean, s the running standard deviation (a prior $\sigma_0 = 1$ until $n \geq 3$), $Q_0 = 0$ a prior value and $\kappa = 1$ the weight of that prior in pseudo-observations. With few observations Q stays near the prior. The count n only narrows the error bar, never inflates the value: an action taken 50 times becomes precisely estimated, not large. This is what prevents the frequency bias of issue #88.

Episodic memory. For the m neighbours of Section 4.3, with weights ω_j and values $v_j = V(G_j \mid H_{now})$:

$$Q_a = \frac{\sum_j \omega_j v_j}{\sum_j \omega_j}, \qquad n_{\text{eff}} = \frac{(\sum_j \omega_j)^2}{\sum_j \omega_j^2}, \qquad SE_a = \sqrt{\frac{\sigma_a^2}{n_{\text{eff}}} + \beta^2}, \qquad \beta = L\,\bar{D}$$

σ_a is the weighted standard deviation of the neighbours' values. n_{eff}, the effective number of neighbours, counts how many equally weighted neighbours the weights are worth: ten close neighbours give $n_{\text{eff}} \approx 10$, one close and nine far give $n_{\text{eff}} \approx 1$. β is a bias term: $\bar{D}$ is the weighted mean distance to the neighbours and $L = 0.5$. Distant neighbours may come from different situations, so their average can be wrong however many there are; β keeps the error bar wide in that case. Sparse, distant or disagreeing memories therefore produce a wide error bar automatically.

World model (deliberation). The JEPA predictor imagines the next latent states for each candidate and the critic scores them. Two changes are needed:

- *What the critic predicts.* Today it predicts an aversive cost (pain + fear), so hunger and sleep outcomes are invisible to it. It should predict the discounted per-drive outcome vector $\hat{G}$, so that $Q_a = V(\hat{G} \mid H)$. In LeCun's architecture (2022) the intrinsic cost is fixed and the trainable critic learns to predict its future value; dl2l's intrinsic cost is the drive deficit d(H). Predicting it is therefore closer to LeCun's design than the current partial cost, not a departure from it.
- *How the planner knows its own uncertainty.* LeCun's architecture does not specify one. We propose a small ensemble of critic heads, whose disagreement gives SE; a head trained to predict outcome variance, which gives σ; and the spread of values across beam-search rollouts, which measures the planner's computational noise. In the simulations the planner's model is a

set of counts, so SE comes from its posterior uncertainty plus a fixed computational noise $\eta = 0.5$, as in Lengyel & Dayan (2007).

Calibration. Each controller's SE is checked against what actually happened. When an episode closes, every decision in it has a realised value, and for the controller that made it we compute

$$z = \frac{V_{\text{realised}} - Q_a}{\sqrt{SE_a^2 + \sigma_a^2}}$$

The denominator is the predictive spread: a single realised outcome differs from Q_a both because Q_a is uncertain (SE) and because outcomes vary (σ). If the controller is honest, z has mean 0 and variance 1. Each controller's SE is multiplied by the root mean square of its recent z (clipped to [0.5, 3]), recomputed during sleep. An overconfident controller thus has its error bars widened.

4.5 Arbitration, budget and exploration

Decisiveness. A controller is decisive when it is confident that its best candidate really beats the second best. With Q_1 and Q_2 the two highest values and SE_1, SE_2 their standard errors,

$$\text{conf} = \Phi\left(\frac{Q_1 - Q_2}{\sqrt{SE_1^2 + SE_2^2}}\right)$$

Φ is the cumulative distribution function of the standard normal distribution. The fraction inside measures the gap between the two best candidates in units of its own uncertainty, and Φ turns it into a probability between 0.5 (the controller cannot tell them apart) and 1 (it is certain). This is the definition of decision confidence used in neuroscience: the probability that the choice is correct given the evidence (Pouget, Drugowitsch & Kepecs 2016). A controller is decisive when conf $\geq \theta = 0.8$.

Budget. Deliberation costs time, and while the creature deliberates its drives keep rising. We reuse the behavioural efficiency of Campos et al. (Section 2.1), normalised to e = E(A)/10, between 0 and 1. The operant table is always affordable; episodic memory requires e $\geq$ 0.3; the world model requires e $\geq$ 0.7. Because E is an inverted U of arousal, the creature deliberates at moderate arousal and falls back on habit when it is calm or close to danger. This matches the finding that acute stress shifts control from goal-directed to habitual behaviour (Schwabe & Wolf 2009), although exact replications of that finding have been inconclusive. An alternative we have not yet tested is to derive the budget from the value of information: escalate when the expected gain of deliberating exceeds the cost of the delay (Pezzulo et al. 2013; Keramati et al. 2011).

The decision.

```
def decide(candidates, H, arousal, dopamine):
    e = efficiency(arousal) / 10                # Campos et al., normalised to [0, 1]
    best, best_conf = None, 0.0
    for ctrl in [operant, episodic, planner]:   # cheapest first
        if e < ctrl.min_efficiency:             # 0, 0.3, 0.7
            break                               # cannot afford to escalate
        est = ctrl.estimate(candidates, H)      # {a: (Q, SE, sigma)}
        conf = decisiveness(est)
        if conf > best_conf:
```

```
            best, best_conf = est, conf
        if conf >= 0.8:
            break                                   # decisive: stop escalating
    if random() < 0.03:                             # small floor of random exploration
        return random_choice(candidates)
    draw = {a: normal(Q, SE * tau(dopamine)) + novelty(a) for a, (Q, SE, _) in best.items()}
    return argmax(draw)
```

Exploration by Thompson sampling. Today dl2l explores through a softmax: `ActionProbabilityFilter` raises each action's probability to the power 1/T, where the temperature T grows with tonic dopamine. A softmax explores the same amount whether the creature knows an option well or not at all. Thompson sampling instead draws, for each candidate, one plausible value from $N(Q_a, (\tau \cdot SE_a)^2)$, a normal distribution centred on the estimate with the controller's own uncertainty as its width, and takes the candidate with the largest draw. Each candidate is chosen with the probability that it is the best one given what the controller knows. Well-known options are rarely re-tried and poorly known ones often, so exploration switches off as evidence accumulates and switches back on when uncertainty returns. Humans use both kinds of exploration, random and uncertainty-directed (Gershman 2018). Tonic dopamine keeps its role by scaling the width through τ.

Three terms complete exploration:

- a **novelty bonus** $0.1/\sqrt{(1 + n_{eff})}$, large when episodic memory has few close neighbours, after the reading of dopamine novelty responses as exploration bonuses (Kakade & Dayan 2002) and the episodic novelty of Never Give Up (Badia et al. 2020);
- a **3% floor of random choices** ($\varepsilon = 0.03$), the device DQN, MFEC and NEC use to guarantee that every option keeps being sampled;
- **tedium**, which dl2l already models as a drive that WANDER relieves, so a bored, sated creature explores by itself.

5 Comparison with Mapa and Campos

The proposal keeps what both predecessors share: selection driven by the attended emotion, stochastic choice, and memory of emotionally valued outcomes. It changes how controllers are combined, what is remembered, and adds deliberation.

Aspect	Campos et al. (2015)	Mapa (2009)	This proposal
Unit of memory	One weight per fruit type	CEQ: ordered sequence of MEEs that restored homeostasis	Engram with key k and discounted outcome vector G, linked to its episode
What is remembered	Accumulated nutritional value, W ← W + Q	Emotional valence of each step; intensity IMR	Per-drive change caused, valued against current need at retrieval
When memory is used	When an affordance tie remains and the object is known	Appraisal picks the stimulus of highest expectancy	Every cycle, with influence proportional to its decisiveness
How controllers are combined	Fixed filter order	Fixed sequence: memory, then operant conditioning	Uncertainty-based arbitration under an efficiency budget

Aspect	Campos et al. (2015)	Mapa (2009)	This proposal
Exploration	Random fall-through when memory is silent	Stochastic evocation of memories	Thompson sampling, novelty bonus, 3% random floor, tedium
Deliberation	None	None	World-model rollouts as an expensive third controller
Episode boundary	None	Regulation of the complex emotion	Attentional focus with hysteresis, plus pre-emption, abandonment and death
Role of arousal	Sets movement speed through E(A)	Sets consolidation through an inverted U	Sets movement speed and which controllers are affordable

Why we expect it to do better. Each change targets a failure seen in dl2l:

- *Fixed order.* With memory first, a creature acts on one or two engrams early in life and cannot be overruled later. Under arbitration, memory's influence grows with its evidence. The 150-interaction hand-off that Campos et al. observed should then emerge as a measured outcome rather than be built in.
- *Value tagged by the wrong drive.* Storing the outcome vector and valuing it against current need fixes issue #91 by construction. Campos's nutritional value Q is the precedent for remembering what an outcome did rather than how the creature felt.
- *No exploration once memory is non-empty.* Thompson sampling keeps exploring exactly the options that memory knows least about.
- *Memory blind to context.* Mapa's expectancy and dl2l's object scores do not depend on the creature's internal state or situation; the kNN key does.

Mapa's CEQ is kept as the storage unit, and her intensity can serve as retrieval accessibility, with importance weighting to keep value estimates unbiased. Valuing outcome vectors rather than scalar valence is a deliberate departure from Mapa and should be reported as such.

Neither predecessor has a world model, so behaviour-parity comparisons (issue #84) should run the arbiter with only the operant and episodic controllers.

Relation to active inference. Without its planning controller, that is with only the operant and episodic controllers, the mechanism maps onto the active inference (free-energy) framework of Friston and colleagues: the set points and the drive-reduction value correspond to prior preferences and pragmatic value, the standard errors to posterior uncertainty about parameters, the novelty bonus to epistemic value, and the decisiveness of a controller to policy precision. Appendix C gives the full correspondence and its limits. The mapping is useful in both directions. It grounds the mechanism in a principle that is not specific to dl2l. It also shows that the mechanism accounts for phenomena that the framework does not describe directly: a limited deliberation budget set by arousal (a Yerkes–Dodson inverted U), the shift from deliberation to habit under stress, a third, episodic controller that sits between habit and planning, and episodes delimited by attentional focus on one drive. These are hypotheses supported so far only by simulation (Sections 6 and 7); Section 8 lists the behavioural tests that would support or refute them.

6 Simulation study: environments and protocol

Before committing cluster time to dl2l, we tested the mechanism in minimal environments where each experiment takes minutes and every quantity is known. The study asks three questions: does arbitration beat the controllers it combines; does the episode marker prevent dithering between drives; and what parameter values should dl2l start from. All code is in NumPy (`arbitration_sims.py`, `multicontext_sims.py`).

6.1 Homeostatic tree environment

The environment extends the tree of Lengyel & Dayan (2007) that we replicated (Section 3.1) with drives. A trip starts at the root; the creature takes two actions, each leading at random to one of three nodes with probabilities fixed per world, and reaches one of 9 leaves. The leaf delivers a relief vector r = −K for (hunger, sleep): food, rest, mixed, empty or harmful. Drives rise by 0.10 and 0.08 per decision and the creature dies at 7. A life lasts 2,000 decisions, and at decision 1,000 the food and rest leaves swap, so what was learned becomes wrong. Worlds were kept only if an agent knowing the true model survives them.

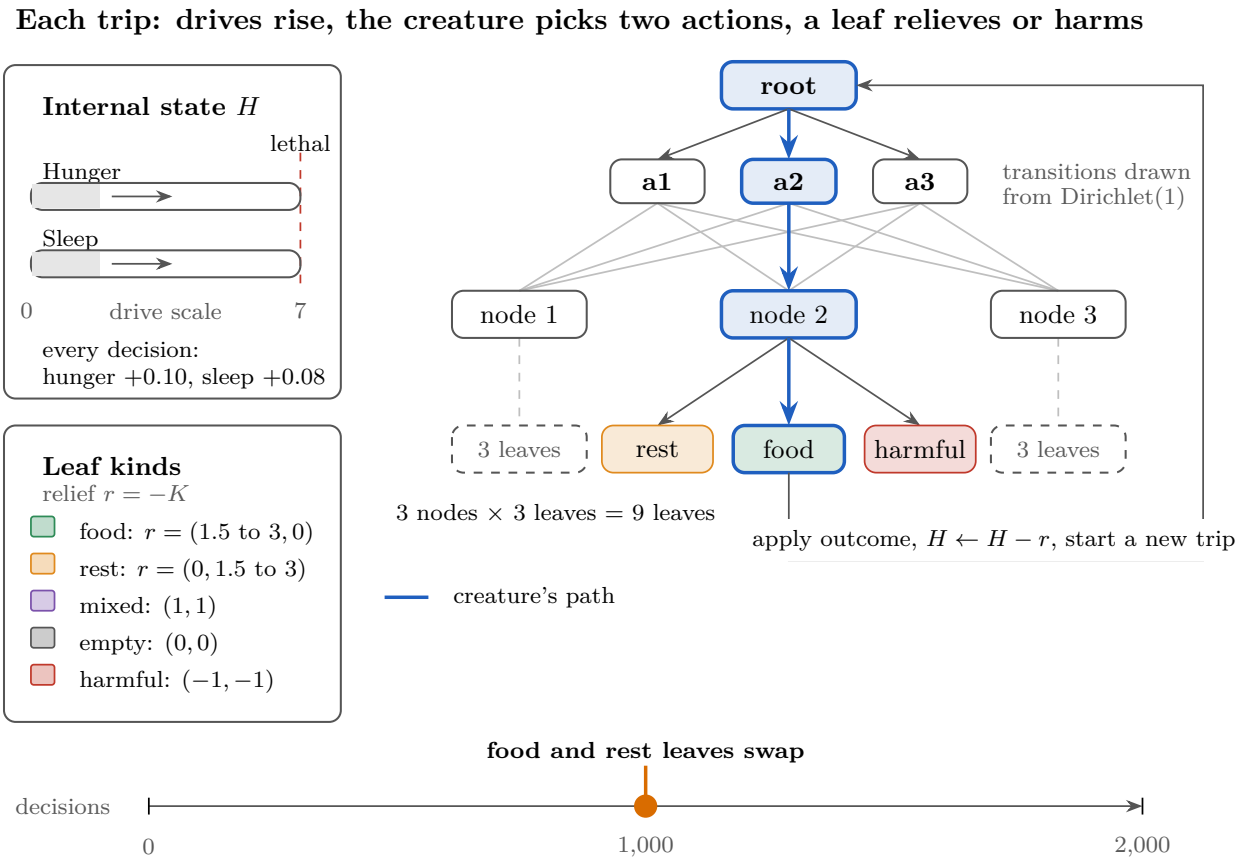


Figure 3. the homeostatic tree environment.

The three controllers are those of Section 4.4. The operant cells are keyed by (dominant drive, node); episodic memory uses the key [node one-hot ; λ·z(H)] with 10 neighbours and 400 engrams per action; the planner keeps transition counts and leaf means and plans by dynamic programming with computational noise $\eta = 0.5$. The arbiter uses $\theta = 0.8$, $\varepsilon = 0.03$, a novelty bonus of 0.1 and the

efficiency thresholds 0, 0.3 and 0.7.

Compared arms: random choice; each controller alone (operant, episodic, planner), each with the same Thompson sampling; a fixed order in which memory decides whenever it has evidence and the operant table otherwise, as in the current dl2l chain; the arbiter; the arbiter without efficiency budget; the arbiter with softmax instead of Thompson sampling; and the arbiter without calibration.

Outcome and statistics. The outcome is the mean drive deficit d(H) over the life, with every step after death counted as 7. Lower is better. All arms run on the same worlds, so comparisons are paired by world, which removes the variation between worlds. We use one-sided Wilcoxon signed-rank tests on the paired differences, report the share of worlds each arm wins, and correct for multiple comparisons with Holm's method. An exploratory run used 40 worlds. A confirmatory run then used 200 new worlds, with hypotheses, tests and the SHA-256 of both scripts fixed in a pre-registration file before any simulation ran.

6.2 Corridor environment

The corridor isolates the episode marker. Food and bed are 10 cells apart; drives rise by 0.06 and 0.05 per cycle, and eating or sleeping at the matching end relieves 0.45 per cycle. Rare surges raise one drive by 1.5 over 30 cycles (5.0 in the stress test). The creature walks to the resource of its focal drive and consumes it. Only the focus rule varies: $\theta_{\text{on}} = 2.0$, $\theta_{\text{off}} \in \{2.0, 1.5, 1.0, 0.5\}$, $\Delta \in \{0, 0.05, 0.15, 0.3, 1.0\}$, with or without the emergency threshold. A rule without hysteresis is the case $\theta_{\text{off}} = \theta_{\text{on}}$ and $\Delta = 0$. Each setting ran 60 lives of 6,000 cycles (100 in the stress test).

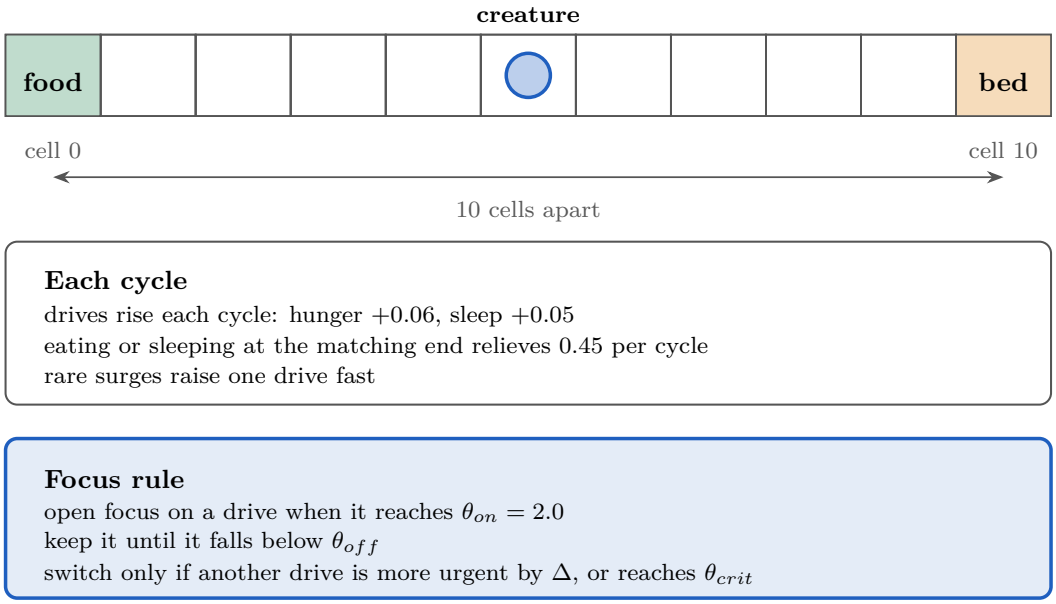


Figure 4. the corridor environment.

6.3 Multi-context environment

In the tree environment one controller turned out to be best almost everywhere (Section 7.1), so arbitration had little to gain. The third environment is built so that different controllers should

win in different situations: seven tree contexts in three families of similar layouts, three drives, one frequent context, rare contexts, contexts that appear mid-life, and a revaluation in which the best leaf of the frequent context becomes harmful. This study is exploratory (40 worlds); its pre-registered run on 200 worlds is pending.

7 Results

7.1 Arbitration in the tree environment

In the pre-registered run on 200 worlds, the arbiter regulated drives better than the planner alone and episodic memory alone, and as well as the operant table alone, the best single controller in this environment. Both pre-registered hypotheses were supported.

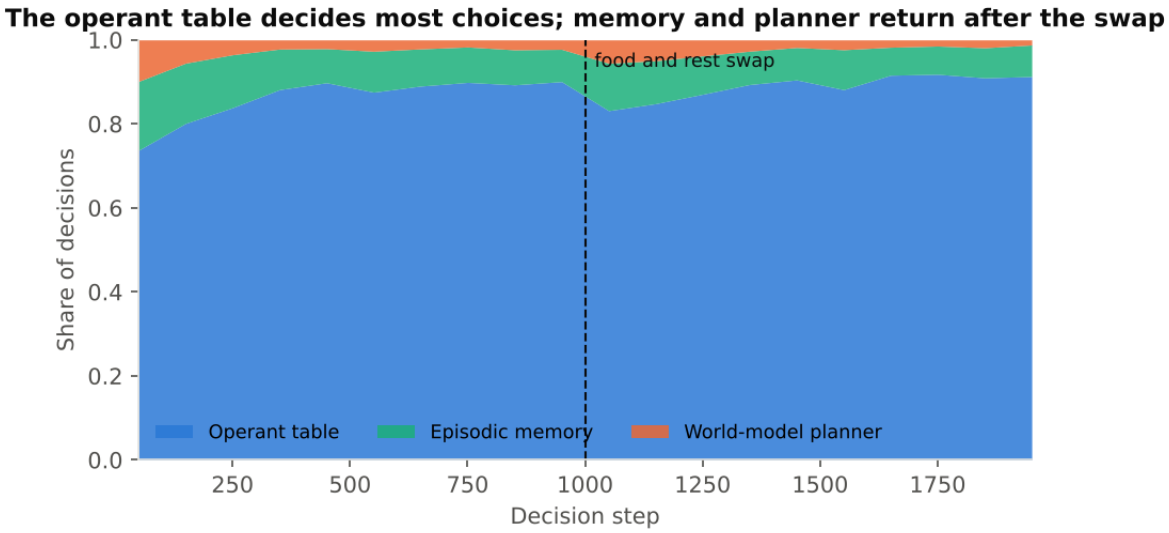
Arm	Mean deficit	Median	Survived 2,000 decisions
Operant table alone	1.73	0.53	76%
Arbiter	**1.75**	**0.55**	**74%**
Episodic memory alone	1.89	0.64	68%
Fixed order (memory first)	1.89	0.64	70%
Planner alone	2.42	0.93	59%

Comparison	Arbiter better in	Median paired difference (95% CI)	Holm-corrected p	Survival
H1: arbiter vs planner	176 of 200 worlds	−0.166 (−0.224 to −0.127)	1.3×10^{-22}	74% vs 59% (McNemar p = 6×10^{-7})
H2: arbiter vs episodic	149 of 200	−0.058 (−0.074 to −0.042)	3.1×10^{-9}	74% vs 68% (p = 0.06)
Arbiter vs fixed order (exploratory)	149 of 200	−0.059 (−0.078 to −0.041)	7.8×10^{-10}	74% vs 70% (p = 0.26)
Arbiter vs operant (exploratory)	95 of 200	+0.007 (−0.007 to +0.013)	0.81	74% vs 76% (p = 0.5)

The gains are reliable but small for a typical world: against episodic memory the median improvement is 0.06 on a median deficit of about 0.6. The arbiter let the operant table decide 81% of choices, episodic memory 13% and the planner 6%. In this environment the operant table is the most reliable controller almost everywhere; the arbiter found that out and deferred to it, without being told. It also beat the fixed order of the current dl2l chain in three worlds out of four.

Over life, episodic memory and the planner decide more of the early choices and more of those just after the food–rest swap, and the operant table takes over as its evidence accumulates (exploratory run, 40 worlds). The direction matches Lengyel & Dayan's analysis, though the operant table dominates sooner than they predict, probably because its cells are coarse and therefore fill quickly.

Ablations (exploratory, 40 worlds). None of the following differences is statistically resolved at 40 worlds; they guide what to test in dl2l.



Share of decisions taken by each controller over life.

- *Exploration.* With Thompson sampling the share of non-greedy choices fell from 43% early in life to 24–26% late, as intended. With a softmax it rose from 41% to 56%. Thompson sampling did not raise exploration again after the swap.
- *Budget.* Removing the efficiency budget changed little (1.53 vs 1.64 mean deficit). Deliberation costs nothing in this environment, so it cannot show the budget's benefit.
- *Calibration.* Without calibration the mean deficit was lower (1.26 vs 1.64) but the calibrated arbiter won in 58% of worlds; unresolved.

7.2 The engram key

Episodic memory alone did best at $\lambda = 0.5$ (mean deficit 2.02), against 2.13 with the percept only ($\lambda = 0$) and 2.66 at $\lambda = 1$. Adding four features of pure noise to the key made it worse at every λ (2.48–2.76). The intervals overlap, but the direction agrees with snake_rl: features that do not predict value hurt the key, and internal state helps when it shapes retrieval without dominating it.

7.3 The episode marker

The switch margin Δ decides whether creatures survive. With $\Delta = 0$ or 0.05 every creature died: they switched focus 7–10 times per 100 cycles and spent 77–80% of their lives walking between food and bed. With $\Delta \geq 0.15$ none died, at any θ_{off}. A lower θ_{off} then regulates better, because each visit consumes more: the mean deficit falls from 3.37 at $\theta_{\text{off}} = 2.0$ to 2.00 at $\theta_{\text{off}} = 0.5$, and walking falls from 67% to 37% of life.

Under large surges the emergency threshold matters:

Focus rule	No emergency threshold	$\theta_{\text{crit}} = 5.5$	$\theta_{\text{crit}} = 4.0$	Time-to-lethal rule
$\theta_{\text{off}} = 0.5$, $\Delta = 1.0$ (commit to the focal drive)	63% die	63% die	0% die	4% die

Focus rule	No emergency threshold	$\theta_{\text{crit}} = 5.5$	$\theta_{\text{crit}} = 4.0$	Time-to-lethal rule
$\theta_{\text{off}} = 1.0$, $\Delta = 0.15$ (switch readily)	14% die	14% die	61% die	35% die

At 5.5 the threshold fired too late for the creature to walk to the other resource. Combined with a small Δ, an emergency threshold added switches and deaths. The configuration that worked was to commit to the focal drive (large Δ) and let an emergency at level 4 override it, applied only while the focal drive itself is below 4; without that last condition two high drives took the focus from each other until the creature died. These values are adopted in Section 4.2.

7.4 Multi-context environment (exploratory)

Three of the four designed situations favoured the intended controller: episodic memory in rare and new contexts, and the planner after revaluation. The operant table, expected to win in the frequent context, was the worst single controller everywhere, because with three drives its cells, keyed by the dominant drive only, were too coarse. Over 40 worlds the arbiter had the lowest mean deficit (2.38, against 2.43 for episodic memory, the best single controller), beating the planner in 36 worlds, the operant table in 32 and episodic memory in 27 ($p = 0.06$). It also beat a hindsight rule that uses the best controller for each situation (27 of 40, $p = 0.04$), which suggests that it gains by mixing controllers decision by decision rather than situation by situation.

One observation is a warning for dl2l: the operant table still made 49–66% of the decisions although it was the least accurate controller. Its cells are confident but biased, and calibration corrects bias only slowly. Decisiveness rewards narrow error bars, not accuracy.

8 Validation plan in dl2l

The aim of dl2l is a creature that behaves like an animal, not only one that survives longer, and Section 7 showed that survival gains may be small. Validation therefore has two parts: an ablation ladder, and tests against behavioural patterns known from animal studies.

8.1 Ablation ladder

Each arm is a pilot (3 trials × 3 creatures) before a 16-trial campaign, sized from the pilot's variance. An arm is adopted only if it meets its criterion; thresholds marked * are to be fixed in the pre-registration.

Arm	Change	Compared with	Adopt if
A0	Current chain, after issues #90 and #93	—	Baseline
A1	Focus tracker replaces the per-cycle most aroused emotion	A0	Fewer focus switches per 100 cycles; median survival not lower
A2	Arbiter with the operant controller only	A1	Regulation not worse*; calibration variance within [0.8, 1.25]*
A3	+ episodic controller, fixed key, $\lambda = 0.5$	A2	Better object choice early in life; survival not lower

Arm	Change	Compared with	Adopt if
A4	Key ablation: $\lambda \in \{0, 0.5, 1\}$; with and without neuromodulators	A3	Best calibration without loss of survival
A5	Same controllers in a fixed order	A3	Arbitration beats the fixed order
A6	+ planner, after the critic is retargeted; budget arms: efficiency, arousal ≥ 4.5, none	A3	Survival or regulation gain; report which budget wins
A7	Emotion-modulated precision weights w and π	best arm	The thesis experiment: does emotion as precision change behaviour as predicted?

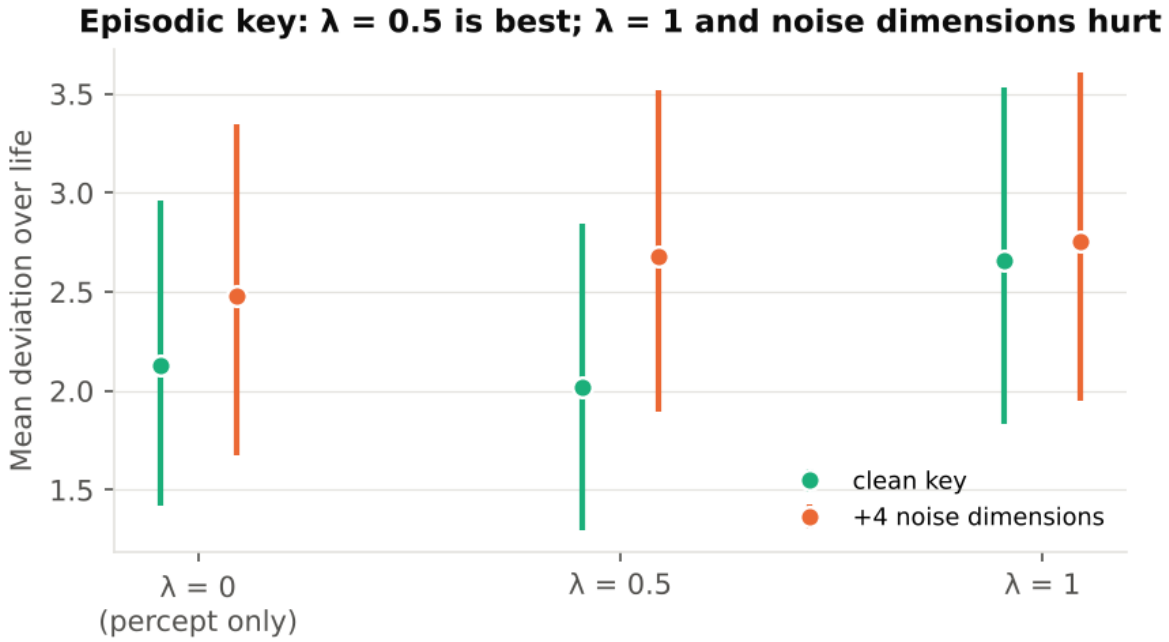
Survival is analysed with Kaplan–Meier curves that account for creatures still alive when a trial ends.

8.2 Comparison with animal behaviour

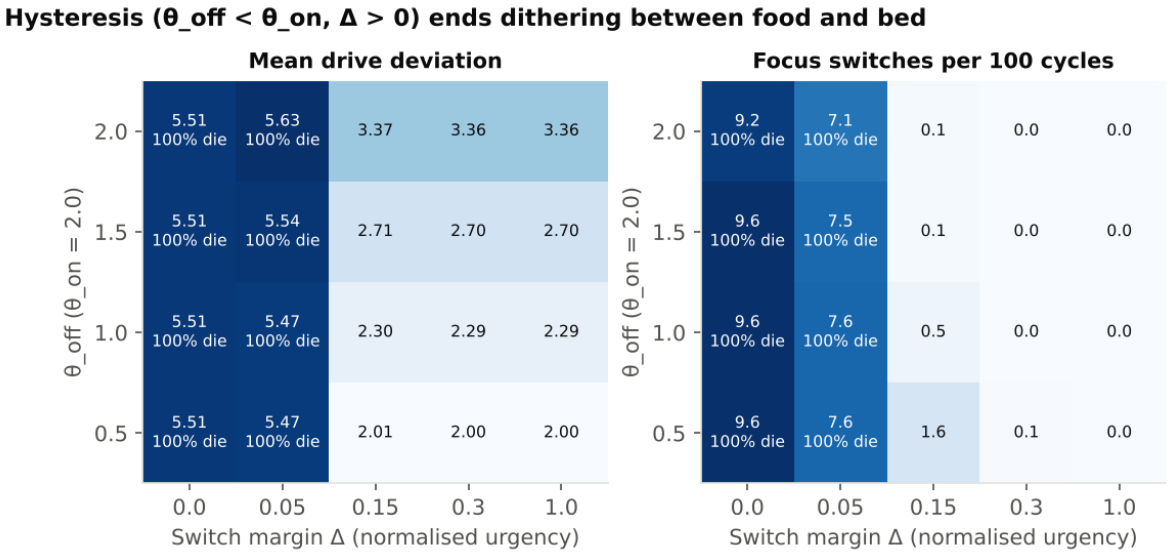
Units differ between a dl2l creature and a rat, so we compare the direction and shape of effects under the same manipulation, as pattern-oriented modelling does for agent-based models in ecology (Grimm et al. 2005). Each paradigm below can be built in the dl2l world, and the arbiter and single-controller creatures predict different results.

Paradigm	What animals do	How to build it in dl2l
Outcome devaluation (Adams & Dickinson 1981)	After little training they stop working for a devalued food; after overtraining they continue (habit)	Make one fruit worthless after short or long exposure; count approaches to it
Stress and habit (Schwabe & Wolf 2009)	Stress makes behaviour insensitive to devaluation	Raise cortisol through the endocrine system before the devaluation test
Patch leaving (Charnov 1976)	Animals leave a depleting patch when its intake falls to the environment's average, later when travel is long	Vary fruit density and the distance between patches; measure time in patch
State-dependent valuation (Pompilio & Kacelnik 2005)	Animals prefer a food met while hungry, even when it pays less	Train two fruits under different hunger; test preference when sated
Speed and deliberation (Keramati et al. 2011)	Deliberate choices are slower, and slowness falls with training	Cycles spent on planner-decided vs habit-decided choices, by age

Four rules keep the comparison fair: predictions are written down before running; control arms (random, single controllers, fixed order) must fail some patterns, otherwise a pattern says nothing about arbitration; parameters are never tuned on the pattern being tested; and patterns that are not reproduced are reported. Devaluation and the stress-to-habit shift come first, because dl2l already has the fruit types, the endocrine system and the neuromodulators they need.



Episodic memory alone by key weighting λ, with and without four noise features.



Corridor: mean deficit and focus switches per 100 cycles by θ_{off} and Δ.

9 Conclusion and open questions

The failure of memory in dl2l comes from how controllers are combined, not only from what memory stores. We propose to combine them by uncertainty: every controller estimates the same quantity, the reduction of the creature's drive deficits, and reports how sure it is; cheaper controllers decide when they are sure, and deliberation is used when it is needed and affordable. Episodes are delimited by attentional focus on one drive, episodic memory works on a fixed hand-built key, and exploration is directed by uncertainty. In minimal environments the mechanism did as well as the best single controller without knowing which it was, beat the fixed order of the current chain, and the episode marker removed dithering between drives.

The mechanism can also be read as active inference without planning (Appendix C). That reading ties the value, uncertainty and exploration terms to a single principle, and it marks what the mechanism adds to it: a deliberation budget set by arousal, a controller for episodic memory, and episodes defined by attentional focus. Whether these additions describe real animals is the question the validation plan of Section 8 is designed to answer.

Open questions:

- Should the budget come from behavioural efficiency or from the value of information? Only an environment where deliberation has a real cost, such as dl2l, can tell.
- How fine should the operant cells be? Too coarse, and a confident but biased habit dominates.
- Does the hand-built key suffice in dl2l, or will the JEPA latent be needed, and can it be kept stable across sleep?
- Will the multi-context result, an arbiter better than every single controller, hold in its pre-registered run?
- Which behavioural signatures does the arbiter reproduce that single controllers cannot?

Appendix A: Integration into dl2l

The integration is staged so that every intermediate build behaves exactly like today until the arbiter is switched on by a flag in `LearningSettings`. Each step follows the project's cycle: an issue, a plan in `docs/plans/`, an experiment specification and a report.

1. **Prerequisites.** Make hunger bind in the test world (issue #90: in the current stack hunger peaks near 1.1 of a lethal 7, so no memory mechanism can show a survival effect), and apply eligibility once instead of twice (issue #93).
2. **Data model.** `Engram` gains the internal-state vector h, the outcome vector K, the key k, the episode identifier and the steps to outcome. `HomeostaticRegulation` attributes K to the stimulus the act produced. New columns go to the Arrow tables and the extraction SQL.
3. **Shared pieces,** in a new package `creature.arbitration`: an `Estimate(Q, SE, σ)` record and a `ValueEstimator` interface; `HomeostaticValue`, computing d(H) and V; `EngramKeyEncoder`, the fixed encoder of Section 4.3, reusing the normalisation statistics of `model_contract.json`; and `KeyMetric`.
4. **Focus tracker.** A `FocusTracker` in `FullAppraisal` owns θ_{on}, θ_{off}, Δ, θ_{crit} and $T_{\max}$, replaces the per-cycle most aroused emotion as the emotion passed to the filters, and emits episode open, close, suspend and resume events.
5. **Operant estimator.** The Welford table of Section 4.4, updated from closed episodes. The existing `OperantConditioning` remains for the parity arms.
6. **Episodic estimator.** In `MemorySystem`: close episodes on the focus tracker's events, compute G for each engram, keep each engram linked to its episode, and answer kNN queries with (Q,

SE, σ) and a novelty signal.

7. **Planner estimator.** Wraps `WorldModelEngine`; requires the critic to predict per-drive outcomes and an ensemble or variance head for SE and σ. Milestone 8's beam search later replaces the one-step score behind the same interface.
8. **`ArbitrationFilter`.** Implements `ActionFilter` with a new `ActionSelectionType`. When enabled, the chain becomes action tendency, nearest target, arbiter. Behavioural efficiency already arrives with each emotional stimulus; tonic dopamine reaches the Thompson width through the existing modulation path.
9. **Telemetry.** A new `ArbitrationDecisionState` per decision: efficiency, the most expensive affordable controller, each consulted controller's decisiveness, the deciding controller, Q, SE and σ of the chosen and runner-up candidates, the source of exploration and the episode. Realised values are joined when the episode closes, which gives calibration and the analyses of Section 8.
10. **Experiments,** as in Section 8.

Steps 2–7 can be unit-tested in isolation; only step 8 changes behaviour. A cheap check is possible before any code changes: `MemoryFilter` already logs the winning and runner-up scores of each decision, from which a decisiveness proxy can be computed on existing campaign data.

Appendix B: Risks

B.1 The mechanism

Risk	Why it could happen	Mitigation
A biased controller dominates	Decisiveness rewards narrow error bars, not accuracy; in the multi-context study the least accurate controller still made half the decisions	Calibration against realised values; keep the operant cells fine enough (focal drive × object × distance); report calibration per controller
A poorly designed key	In snake_rl, MFEC fell to chance when the key held features unrelated to value; neuromodulator and hormone levels may play that role in dl2l	Standardise features, weight blocks equally, ablate λ and the feature set; later, set the precision weights π_i from how well each feature predicts outcomes
Tuning until it wins	The mechanism has about a dozen constants; NEC's published advantage disappeared once its baseline was tuned equally	Fix most constants by principle; tune only θ, λ and ε, on held-out seeds, with the same budget for the baseline chain; pre-register
A controller never calibrates	A controller that never wins never sees its predictions tested	Log every affordable controller's estimate for the chosen action; run the planner in the background on a fraction of cycles
The arousal loop	High arousal → low efficiency → habits only → poorer regulation → higher arousal	Biologically plausible, but it must not dominate mortality; report deaths by arousal state
A fixed budget is wrong for dl2l	Planning has a real cost in dl2l (inference time, cycle rate), unlike in the simulations	Compare the efficiency budget with no budget and with the current arousal $\geq$ 4.5 gate as experimental arms

Risk	Why it could happen	Mitigation
Small effects	In the tree study the typical gain was 0.06 on a deficit of 0.6	Measure where controllers differ in reliability (Section 8), not only survival over the whole life

B.2 Interaction with the JEPA world model

Risk	Why it could happen	Mitigation
Stale keys	If the JEPA latent became the memory key, every sleep-time adapter update would move the latent space and misplace older engrams	Keep the fixed encoder as the key; if the latent is adopted later, re-encode stored keys during sleep from the stored percept and internal state
Training through memory	NEC's failures: embedding explosion from constant inputs and dependence on reward scale	No gradient from memory values into the encoder; any learned key is trained on its own objective, behind the existing collapse check
The latent ignores value	In snake_rl every agent's embedding was organised by position, which hid the food from neighbour lookups	Before using the latent as a key, test whether it separates object type and dominant drive
The critic predicts the wrong quantity	It scores pain and fear only, so the planner cannot value food or sleep	Retarget the critic to per-drive outcomes (Section 4.4); until then, keep the planner out of feeding and sleeping decisions
Distribution shift	The species model was trained on trajectories from the old selection rule; the new policy visits other states, error rises and the planner stops being consulted	Background evaluation; retrain the adapter during sleep on post-arbiter data; keep the out-of-distribution gate
Latency	Beam search over about eight actions per cycle; DJL predictors are not thread-safe	Escalate only when cheaper controllers are indecisive; one predictor per creature or a pool; monitor cycle rate (issue #85)

Appendix C: Correspondence with active inference

This appendix maps the mechanism, without its planning controller, onto active inference (Friston et al. 2016; Parr, Pezzulo & Friston 2022 for the textbook account). In that framework an agent selects policies π with low expected free energy $G(\pi)$, which combines a pragmatic term (reaching preferred outcomes) and an epistemic term (resolving uncertainty). The correspondence below is an interpretation, not a derivation; the last column says how close each match is.

Active inference	This mechanism	Fit
Prior preferences over outcomes, p(o)	Set point H* and deficit d(H)	Exact only if d is $-\log$ p(H) under Gaussian preferences, which needs the squared distance; d(H) as defined is a distance, so the match is approximate
Pragmatic value, expected log-preference of outcomes	Value V = d(H) − d(H+K), the reduction of drive deficits	Close; V is the change in the negative log-preference when d is squared
Precision of preferences	Drive weights w (the emotion-as-precision hypothesis)	Analogous (Pezzulo, Rigoli & Friston 2018; Hesp et al. 2021); w = 1 in this proposal
Posterior uncertainty about model parameters	Standard error SE of each controller	Close
Epistemic value, information gain	Novelty bonus and Thompson sampling	Same role, different route: AI scores the expected information gain of each policy, we sample from the posterior (Schwartenbeck et al. 2019)
Ambiguity, expected uncertainty of outcomes given states	Outcome variance σ	Partial: AI avoids ambiguous policies, we use σ only to calibrate SE
Policy precision γ, the inverse temperature of policy selection, linked to dopamine (Friston et al. 2012)	Decisiveness conf and the dopamine temperature of ActionProbabilityFilter	Analogous: both set how deterministic the choice is
Habit prior E(π), learned from repeated choices	Operant table	Close (Friston et al. 2016)
Posterior over policies, P(π) $\propto$ exp(ln E − γG), a precision-weighted blend of habit and planning	Arbiter that selects one controller by decisiveness	Different: AI blends, we select (FitzGerald, Dolan & Friston 2014); see the variant below
Learning of precisions (hyperpriors)	Calibration scale z, clipped to [0.5, 3]	Analogous
No counterpart	Episodic controller on a hand-built key	Extension: a third controller between habit and planning
No counterpart	Efficiency budget e = E(A)/10 gating deliberation	Extension: the nearest idea is bounded rationality, where information-processing cost limits deliberation (Ortega & Braun 2013)
No counterpart	Episodes delimited by attentional focus on one drive, with hysteresis	Extension: AI does not segment experience by the regulation of a drive

What the mapping supports. With the operant and episodic controllers only, value, uncertainty, exploration and confidence each have a counterpart in active inference. The mechanism is therefore compatible with the principle and does not need a separate justification for those terms.

What the mechanism adds. Three components have no counterpart: the arousal-dependent budget, the episodic controller, and the drive-focused episode. Together they predict phenomena the framework does not describe directly, including the shift from deliberation to habit under stress, the cost of deliberating while drives rise, and the choice among controllers of different cost. Each is testable with the signatures in Section 8.2.

Limits. The squared-distance caveat above means the value is not literally a free-energy term. The mechanism selects one controller where active inference blends them. The planner, which would be a model-based generative model in active inference terms, is outside this mapping.

Variants to test. (1) Precision-weighted blending: mix the controllers' value estimates with weights $1/\mathrm{SE}^2$ instead of selecting one, as an extra arm in the simulations of Section 6. (2) Ambiguity penalty: subtract a multiple of σ from Q, which should make the creature prefer reliable outcomes, a signature of risk-sensitive choice. (3) Squared deficit: replace d(H) by its square and check whether the results of Section 7 change. If blending matches or beats selection, the budget would be the only part of the arbiter that active inference does not already cover.

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Project sources: snake_rl study summary; dl2l issues #84, #88, #90, #91, #93 and PRs #89, #92; dl2l — Memory in Action Selection: Review & Design (September 2026).